

Shubham Kaushik

Data Systems Researcher | Software Engineer

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RESEARCH INTERESTS

Database Systems, Storage Engines, Data-intensive Infrastructure, and Learning-based System Tuning

EDUCATION

Doctor of Philosophy | Brandeis University

Major: Computer Science (GPA: 3.98/4.0)

Jan 2024 - Present

Waltham, MA, USA

Master of Science | Boston University

Major: Computer Science with specialization in "Data-Centric Computing" (GPA: 3.88/4.0)

Sep 2022 - Dec 2023

Boston, MA, USA

Bachelor of Technology | Maharshi Dayanand University

Major: Computer Science & Engineering

Jul 2014 - Jun 2018

Haryana, India

Thesis: "Fault Modelling of an Object-Oriented System using Colored Petri Nets"

EXPERIENCE

Ph.D. Researcher, Smart & Scalable Data Systems Lab @ Brandeis University, MA, USA

Jan 2024 - Present

- Design and evaluate LSM-based storage engine internals such as merge policy, layouts, and write-buffer design.
- Apply learning-based methods to storage-engine configuration, from offline tuning to runtime adaptation.
- Develop benchmarking suites used to evaluate key-value stores under dynamically shifting workloads.
- Built a Domain-Specific Language that restrict LLMs to generate eight edit commands to produce valid benchmarks.

Software Engineer, Scalability & Infrastructure Team @ Kwalee, India

Mar 2022 - Aug 2022

Engineer - Information Security, Information Security Automation @ FIS Global, India

Jun 2021 - Mar 2022

- Built an asynchronous IP-scanner to scan over 100K+ internet endpoints, replacing the team's manual spot-checks.
- Authored and merged integrations for DB2, GenericSQL and FireMon into Palo Alto's Cortex XSOAR [[demisto/content](#)].

Project Engineer, Cloud Computing and Big Data Teams @ Wipro Limited, India

Jul 2018 - Jun 2021

- Built a microservice that pseudonymises PII in CSV, Excel and mainframe files, replacing a manual review of 1K+ files.
- Designed a real-time agent that flags anomalous hosts across 500+ enterprise systems using k-means clustering.

TECHNICAL SKILLS

Programming Languages: C, C++, Python, Rust

Databases: RocksDB, PostgreSQL, Redis

Systems & Performance: Linux, perf, gdb, CMake, Docker

Machine Learning: Reinforcement Learning (RL), LLMs, DSL, Retrieval-Augmented Generation (RAG), PyTorch

PUBLICATIONS

- Mind the Buffer: Dissecting the LSM-Buffer Design Space* [[blog](#)] [[GitHub](#)] (In Review)
Boao Chen, **Shubham Kaushik**, and Subhadeep Sarkar
- Tectonic⁺⁺: A Resource-Aware Key-Value Benchmarking Suite for Shifting Workloads* (In Review)
Boao Chen, **Shubham Kaushik**, and Subhadeep Sarkar
- TexBench: A Unified Benchmarking Suite for Shifting Workloads* (demonstration) [[blog](#)] [[PDF](#)] [[GitHub](#)] VLDB 2026
Abhishek Chanda, **Shubham Kaushik**, Artem Lavrov, and Subhadeep Sarkar
(52nd International Conference on Very Large Data Bases)
- TexBench: Harnessing LLMs for Efficient Key-Value Benchmarking* [[blog](#)] [[PDF](#)] TPCTC 2026
Shubham Kaushik, Abhishek Chanda, and Subhadeep Sarkar
(18th TPC Technology Conference on Performance Evaluation & Benchmarking)
- RangeReduce: Query-Driven Compaction In LSM-Trees* [[talk](#)] [[PDF](#)] [[GitHub](#)] ICDE 2026
Shubham Kaushik, Manos Athanassoulis, and Subhadeep Sarkar
(42nd IEEE International Conference on Data Engineering)
- Tectonic: Bridging Synthetic and Real-World Workloads for Key-Value Benchmarking* [[PDF](#)] [[GitHub](#)] TPCTC 2025
Alexander H. Ott, **Shubham Kaushik**, James Chen, and Subhadeep Sarkar
(17th TPC Technology Conference on Performance Evaluation & Benchmarking)
- Anatomy of the LSM Memory Buffer: Insights & Implications* [[PDF](#)] DBTest 2024
Shubham Kaushik and Subhadeep Sarkar
(In Proceedings of the International Workshop on Testing Database Systems)
- Fault Modelling of an Object-Oriented System using CPN* [[PDF](#)] JCSE 2019
Shubham Kaushik and Ratneshwer
(International Journal of Computer Sciences and Engineering)

ONGOING WORK

FluidLSM — self-tuning storage engine [blog]	2025 – Present
<i>Designing an LSM engine that retunes itself as workloads shift, using RL and LLM-guided retrieval to search its configuration space.</i>	
Learned-LSM Buffer — workload-aware write buffer	2026 – Present
<i>Developing a learned memory buffer that predicts workload shifts and switches its implementation to hold a performance target.</i>	

TALKS [SELECTED]

<i>RangeReduce: Query-Driven LSM Compactions @ MiDAS Seminar, Boston University, USA (<i>invited</i>)</i>	04/2026
<i>Tectonic: Bridging Synthetic and Real-World Workloads @ 2026 New England Systems Day, Harvard, USA</i>	01/2026
<i>Range Queries and LSM-Trees: From Foes to Friends @ Salesforce, USA (<i>invited</i>)</i>	10/2025
<i>Advancements in LSM Buffering: Performance and Implementations @ Brandeis University, USA</i>	09/2024

POSTERS

<i>Tectonic: Bridging Synthetic and Real-World Workloads @ 2026 New England Systems Day</i>	01/2026
Alexander H. Ott, Shubham Kaushik , Boao Chen, and Subhadeep Sarkar	
<i>RangeReduce: A Range Query Driven Compaction for LSM-Trees @ NEDB Day [PDF]</i>	2025, 2024
Shubham Kaushik , Manos Athanassoulis, and Subhadeep Sarkar	

PROFESSIONAL SERVICE

Program Committee Member, VLDB 2027	2026
Member, VLDB 2026 Shadow PC	2026
Web Chair for Northeast Database (NEDB) Day 2025	2025
External reviewer for IEEE International Conference on Big Data	2025, 2024
Member, ACM and IEEE	2024 – Present

TEACHING [ASSISTANT/GRADER]

Fundamentals of Computer Systems (COSI 131A), Brandeis University [FA '26]	Fall 2026
Database Management Systems (COSI 127B), Brandeis University [SP '26] [SP '25] [SP '24]	Spring 2026, 2025, 2024
Introduction to Computer Networking (COSI 128A), Brandeis University	Fall 2025, 2024
Data Mechanics (DS 310), Boston University	Fall/Spring 2023
Computer Networks (CS 455), Boston University	Fall 2022

OUTREACH

Volunteered at VLDB 2026 , managing the event and assisting session chairs	Sep 2026
Judged and mentored at HackMIT , aiding teams with technical challenges	Sep 2025, 2023
Mentored 4 teams, with an average of 20 participants at BostonHacks	Nov 2022